

PORTFOLIO #2: CONDITIONALS

CONCEPT: SPECIATION ANALYZER

In an analytical chemistry calculation, the state of an acid (protonated/deprotonated) can be determined based on the relationship between the pH of the solution and the pKa of the chemical.

This code uses conditional statements to decide which species dominates.

```
In [17]: def check_chemical_speciation(pH, pKa):  
    # Calculate the difference to determine the dominant species  
    diff = pH - pKa  
  
    print(f"Current pH: {pH}")  
    print(f"Substance pKa: {pKa}")  
    print("-" * 30)  
  
    # Conditionals: Logic based on Analytical Chemistry Calculations  
    if diff <= -1:  
        # pH is at least 1 unit below pKa (mostly protonated)  
        return "DOMINANT SPECIES: Acidic form [HA]. Logic: Protonated (>90%)."  
  
    elif diff >= 1:  
        # pH is at least 1 unit above pKa (mostly deprotonated)  
        return "DOMINANT SPECIES: Basic form [A-]. Logic: Deprotonated (>90%)."  
  
    else:  
        # pH is near the pKa (buffer region)  
        percentage_a_minus = (10**diff / (1 + 10**diff)) * 100  
        return f"STATUS: Buffer Region. Mixture of forms. [A-] is approx {percentage_a_minus}%"  
  
    # Simulation: Checking Acetic Acid (pKa ~4.76) in a solution of pH 6.0  
    print(check_chemical_speciation(pH=6.0, pKa=4.76))
```

Current pH: 6.0

Substance pKa: 4.76

DOMINANT SPECIES: Basic form [A-]. Logic: Deprotonated (>90%).